Xin XIONG

lucimorl123@gmail.com | 17340684873 | xiongxin.online | github.com

Education

Sichuan University, Life Science — Sept 2023 – Jun 2027

- GPA: 3.89/4.0 (transcript)
- Rank: 1/65
- **Selected coursework:** Genetics (96), Organic Chemistry (96), Calculus (94), Microbiology (91), Cell Biology (90), Biochemistry (89)

Honor and Grants

- National Scholarship (2024 & 2025; approximately top 0.3% nationally)
- iGEM Gold Medal & Undergraduate Top 10 Project (2025)
- Outstanding student of Sichuan University

Projects

Pan-Cancer Analysis of Tumor Microenvironment Cell-Communication Networks — Jun 2025–Present

- **Data integration:** Integrated ∼4.7 million single-cell transcriptomes across 30+ cancer types to characterize recurrent tumor-microenvironment cell states and multicellular communication programs.
- **Module discovery:** Combined ligand–receptor inference and NMF-based module discovery to prioritize conserved immune- and stromal-associated regulatory niches across cancer cohorts.
- **Spatial validation:** Mapped selected communication programs onto spatial transcriptomic datasets to evaluate in situ sender–receiver colocalization and spatial niche organization.

EMT and Altered KRAS Inducibility in KRAS-Independent Pancreatic Cancer — Jul–Aug 2026

- **Transcriptomic profiling:** Compared parental AK38 and recurrent 772 PDAC cells across Dox ON/OFF conditions using DESeq2, GSEA, and pathway inference; identified EMT and E2F/G2M/MYC programs.
- **Cross-dataset analysis:** Reanalyzed three public KRAS-withdrawal and relapse datasets to examine the reproducibility and specificity of EMT-associated transcription.
- **Proteomics & controls:** Integrated RPPA, immunoblotting, genotyping, and a non-targeting CRISPR control to assess KRAS inducibility and prioritize DNMT1 and selective PDK1/AKT-related signals.

TP53RK Functional-Mode Analysis: Phosphorylation vs. KEOPS Dependence — Jan–Feb 2026

- **Mutagenesis:** Constructed phosphorylation-site and KEOPS-complex-site mutants of TP53RK and introduced them by lentiviral transduction.
- **Functional rescue:** Degraded endogenous TP53RK with dTAG-V1 and used Incucyte to compare proliferation and cell-death rescue between mutant classes.

iGEM SCU-China 2026: DunaTerp – Team Captain — Jun-Oct 2026

- **Leadership:** Led an interdisciplinary iGEM team, coordinating research planning, wet lab, dry lab, hardware, Human Practices, and project delivery.
- **Scientific strategy:** Directed DunaTerp, a *Dunaliella salina* terpenoid platform exploring TF2146 regulation of beta-carotene biosynthesis and downstream product designs.

iGEM SCU-China 2025: TasAnchor Whole-Cell Immobilization Platform — Jun 2024–Oct 2025

- **Recognition:** Contributed to an iGEM Gold Medal, Undergraduate Top 10, and Best Bioremediation Project recognition in 2025.
- **Platform development:** Engineered a *B. subtilis tasA* knockout with a SpyTag/SpyCatcher adhesion scaffold; validated biofilm immobilization on bioreactor membranes.

Experiences

University of Melbourne Visiting Honours Student - WEHI — Sep–Now 2026
University of Notre Dame Summer Visiting Student - Xin Lu Lab — Jul–Aug 2026
Beutler Institute Exchange Student — Mar–Jun 2026
Westlake University Visiting Student - Jin Xin Lab — Jan–Feb 2026

Technologies

Wet lab: PCR, Southern blot, Western blot, plasmid assembly, CRISPR, protein expression and purification, cell culture
Computational & bioinformatics: Python, R, Linux, machine learning, scRNA-seq, spatial transcriptomics, CellChat, NicheNet, CellPhoneDB, BLAST